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Quantitative analysis of the olive oil adulteration based on the assessment of multi-angle LIF spectral data using learning algorithms☆

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ABSTRACT

This study presents a comprehensive assessment of the laser-induced fluorescence (LIF) spectroscopy for detecting olive oil (OO) adulteration with sunflower oil (SO), addressing the challenges posed by secondary inner filter effect (sIFE). Various machine learning models, including linear discriminant analysis (LDA) and support vector machine (SVM), as well as deep learning models including multilayer perceptron (MLP) and one-dimensional convolutional neural network (1D-CNN), were evaluated for quantitative classification of different adulteration levels. Novel cross-validation strategies were implemented to assess the generalizability of the models to comprise the influence of the sIFE on LIF spectra across various detection angles. Notably, models trained on multi-angle LIF data demonstrated consistent, near-perfect accuracies when evaluated with data from arbitrary detection angles. Through eliminating the limitations of having to use specified detection angles, our methodology simplifies the use of LIF spectroscopy as a tool for evaluating the authenticity and quality of the edible oils.

1. Introduction

Olive oil, renowned for its nutritional and medicinal benefits, holds a significant place in the culinary and agricultural sectors. However, the high value and demand for authentic olive oil (OO) have led to an increase in incidents of economically motivated adulteration, often involving cheaper oils such as sunflower oil (Klinar et al., 2024). Adulteration not only diminishes the quality and health benefits of OO but also deceives consumers and undermines market integrity. Therefore, the precise and quantitative detection of OO adulteration is critical for ensuring product authenticity, consumer protection, and fair trade practices. Quantitative analysis of OO adulteration presents substantial challenges due to the complexity of the oil's composition and the subtle differences introduced by adulterants. Advanced analytical techniques are essential for detecting and quantifying these differences accurately.

Conventional methods in the classification and quality control of edible oils include classical chemical analysis (Monfreda et al., 2014), gas chromatography mass spectrometry (GC-MS) (Sun et al., 2015, Troya et al., 2015), high-performance liquid chromatography (HPLC) (Khursheed et al., 2024), and Differential Scanning Calorimetry (DSC) (Mihociu et al., 2019, Mahajan et al., 2025). These techniques, while offering high sensitivity and efficiency, often require extensive sample preparation and involve the use of large and complex equipment, making them time-consuming and costly. Classical chemical analysis and chromatographic techniques focus on the quantitative detection of specific chemical compositions; therefore, these methods can be limited in their ability to provide comprehensive information about oil samples. Despite the development of electronic nose technologies for detecting

☆ This study introduces a novel multi-angle Laser-Induced Fluorescence (LIF) spectroscopy framework integrating machine learning (SVM, Random Forest, XGBoost, LDA, PLS-DA) and deep learning models (1D-CNN, MLP) to leverage inner filter effect (IFE) distortions as features of geometric variability for olive oil authentication. Departing from conventional IFE correction approaches, we harness angle-dependent spectral patterns (20°–90° detection) through multi-angle data acquisition, eliminating mathematical/instrumental corrections. Our methodology combines Difference Spectra Analysis (DSA), ANOVA, and PCA with advanced machine/deep learning models to exploit spatial anisotropy in secondary IFE, achieving 100 % classification accuracy using SVM and 1D-CNN via two novel validation protocols: Single-Angle Cross-Validation (SACV, testing angular generalization from single-angle training) and Leave-One-Angle-Out Cross-Validation (LOAOCV, assessing multi-angle robustness). The 1D-CNN demonstrated perfect accuracy (100 %) in LOAOCV testing across unseen angles, while SACV revealed limitations in angular generalization from single-angle training. This paradigm shift unifies multi-angle spectroscopy with intelligent feature extraction, enabling portable adulteration detection systems resilient to geometric variability without requiring spectral preprocessing.

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OO adulteration (Majchrzak et al., 2018, Rodrigues et al., 2019, Haddi et al., 2013) their use is constrained by issues such as poor selectivity, low sensitivity, and signal noise. Spectroscopic techniques, such as Fourier Transform Infrared (FTIR) (Gagour et al., 2024) Near-Infrared (NIR) (Du et al., 2021, Ferreiro-González et al., 2018) and Raman spectroscopy (Wu et al., 2023), have long been utilized for the quality assessment of food products, including edible oils. These methods are valued for their rapid, non-destructive analysis capabilities, offering comprehensive chemical profiles by capturing unique spectral fingerprints. FTIR and Raman spectroscopy, for instance, have proven effective in classifying and authenticating oils, although they often require extensive sample preparation and large, complex equipment, which can hinder their practicality for routine use. Given these constraints, there is an increasing demand for analytical techniques that are not only rapid and cost-effective but also portable and less reliant on extensive sample preparation.

Laser-Induced Fluorescence (LIF) has emerged as a promising alternative, addressing many of these limitations while providing high sensitivity and specificity owing to the utilization of a laser light source at a specific wavelength. LIF spectroscopy captures detailed spectral information on the molecular composition of oils, making it particularly effective for detecting minor adulterations. LIF provides rapid response, high sensitivity, and selectivity. Compared to traditional fluorescence methods using xenon lamps, mercury lamps, or LEDs, LIF significantly enhances fluorescence efficiency and detection sensitivity through its laser source (Szilagyi et al., 2018). The technique requires minimal sample preparation and is non-destructive, making it both cost-effective and environmentally friendly. LIF's ability to provide comprehensive spectral data in real-time, coupled with its high sensitivity to changes in the molecular environment of fluorescent molecules, makes it particularly suitable for detecting subtle alterations in OO composition. These characteristics position LIF as an ideal tool for routine quality control and real-time monitoring in the food industry.

The integration of spectroscopic techniques with machine learning and deep learning algorithms has significantly enhanced the analysis and classification of complex spectral data, particularly in detecting edible oil adulteration. Principal Component Analysis (PCA) serves as a powerful tool for dimensionality reduction, transforming large datasets into principal components that highlight variations due to adulteration, as demonstrated in studies (Farres et al., 2023, Srata et al., 2023). Recent studies have also demonstrated the effectiveness of various machine learning and deep learning combined with spectral data from spectroscopic techniques, including laser-induced breakdown spectroscopy (LIBS) (Nanou et al., 2023), LED-induced fluorescence spectroscopy (Zhang et al., 2022), excitation-emission matrix fluorescence (EEMF) spectroscopy (Chen et al., 2023), and FTIR-HATR spectroscopy (De Santana et al., 2019).

LIF spectroscopy has been particularly successful when combined with machine learning techniques. LIF spectroscopy has been employed in conjunction with chemometric techniques to classify cocoa beans as organic or conventional, achieving up to 100 % classification accuracy with models such as Linear Discriminant Analysis (LDA), SVM, NN, and Random Forest (Pappoe et al., 2023). Similarly, Various vegetable oils were characterized and distinguished using LIF with a 532 nm laser by combining LIF with PCA and SVM algorithms, achieving a recognition rate of up to 100 % for ten popular edible oils (Mu et al., 2013). Adulteration of extra virgin olive oil with rapeseed and peanut oils was additionally detected and quantified using 473-nm LIF, achieving a classification accuracy of 100 % with errors lower than 2 % through artificial neural network (ANN) and SVM algorithms (Mu et al., 2016). LIF spectroscopy and a 1-dimensional convolutional neural network (1D-CNN) with a dual convolution structure were used to classify olive oil, achieving a high accuracy of about 99.69 % with LIF spectra data sets (Chen et al., 2022). LIF spectroscopy has proven effective in analyzing plant-based materials, revealing optical properties and molecular compositions. It has been used to study Echium amoenum (Rahmatpanahi and Bavali, 2025), Malva sylvestris (Bavali et al., 2024), black seed oil (Rahmatpanahi and Bavali, 2024), and sour cherry kernel oil (Rahmatpanahi et al., 2024), successfully detecting adulteration and contamination. In food safety, LIF has achieved 97 % accuracy in identifying Aspergillus molds and AFB1 in rice (Wang et al., 2023) and advanced methods have enhanced fluorescence recovery for AFB1 detection in oils (Wang et al., 2025). Combining LIF with machine learning models has further improved adulteration detection, achieving up to 99.06 % accuracy for avocado oil (Bavali et al., 2025).

In conventional fluorescence setups, the accuracy of selective spectroscopy for specific species is limited by the non-uniform absorption of the excitation light (primary inner filter effect) and the self-absorption of the fluorescence emission (secondary inner filter effect)(Kimball et al., 2020). These effects lead to spatial heterogeneity in fluorescence signals, causing spectral data to vary with the detection angle. In a recent study, the angular dependence of the secondary inner filter effect (sIFE) in the authentication and adulteration detection of OO was extensively studied (Bavali and Rahmatpanahi). Researchers have considered IFE as an undesirable feature and have devoted considerable effort to developing mathematical and instrumental methods to mitigate or correct IFE-induced distortions (Panigrahi and Mishra, 2019). While these correction methods have shown promise, they often face limitations when applied to optically dense, heterogeneous samples containing multiple fluorophores, significantly reducing the practicality and effectiveness of traditional IFE correction techniques in these complex scenarios (Ceresa et al., 2021).

Machine learning and deep learning have emerged as advanced tools in fluorescence spectroscopy data analysis, offering cutting-edge solutions for complex sample evaluation. Despite their potential, there has been no study to date that explores the application of these computational methods in developing robust analytical frameworks capable of accounting for IFE-dependent variations in LIF spectra. Specifically, the challenge lies in maintaining predictive accuracy when analyzing spectral data influenced by varying or previously unencountered IFE distortions. While earlier research has employed the LIF technique alongside chemometric methods for adulteration detection, a comprehensive comparative analysis of machine learning and deep learning models for quantitative evaluation using LIF spectral data remains absent. Furthermore, the neglect of IFE—an inherent source of noise that alters fluorescence spectra—can drastically compromise the performance of classification models. Addressing this gap requires a reliable and generalized methodology that ensures high classification accuracy across diverse IFE contributions, including unseen distortions. This study introduces and critically evaluates such a methodology, aiming to overcome current limitations in LIF spectroscopy and enable the development of simplified and flexible implementations that alleviate constraints in detection setups. By enhancing the robustness and applicability of LIF spectroscopy, this work seeks to advance its use in the analysis of complex samples.

2. Materials and method

2.1. LIF setup

The experimental setup for Laser-Induced Fluorescence (LIF) spectroscopy used in this study followed the same arrangement described in (Bavali and Rahmatpanahi). Oil samples were individually placed in cylindrical cuvettes with an inner diameter of 20 mm and wall thickness of 1 mm. A continuous-wave diode laser, emitting at a wavelength of 405 nm, was employed to illuminate the samples in a cylindrically symmetric configuration. Fluorescence emission was captured using a collimator and subsequently detected by the fiber probe of an OPCT-V550 spectrometer, offering a spectral resolution of 0.6 nm. To examine the angular dependence of LIF spectral features, the fiber probe was mounted on a rotatable arm positioned in a fixed proximity to the cuvette surface.

2.2. Sample preparation

Virgin olive oil (VOO) and Sunflower oil (SO) from reputable brands were sourced from a local market in Gilan Province, Iran. The suppliers guaranteed the quality grades of these oils, which also met the certification requirements of the Institute of Standards & Industrial Research of Iran for authentic edible oils. The samples were checked to ensure they were fresh and had production dates within acceptable limits.

The pure samples consisted of VOO and SO. To prepare the adulterated samples, SO was added to VOO samples with volume ratios of 1:4 (20 % adulteration), 1:2 (33.3 % adulteration), 3:7 (42.8 % adulteration), and 1:1 (50 % adulteration). In this study, the quantitative analysis and classification scheme includes six categories ranging from 0 % (representing pure VOO) to 100 % (representing pure SO), with intermediate percentages indicating various specific volume ratios of SO present in oil compositions (referred to as adulteration percentage). The measurements for the VOO and SO samples were repeated 20 times at eight different angles (20°, 30°, 40°, 50°, 60°, 70°, 80°, and 90°) with respect to the laser beam direction, yielding approximately 160 LIF spectra per pure sample. The adulterated samples underwent 30 repetitions of measurement each, also at the same eight angles, yielding 240 LIF spectra per adulteration level. In total, 1278 LIF spectra were generated, providing a comprehensive dataset for analyzing the spectroscopic characteristics of VOO adulteration with SO.

2.3. Data preprocessing

The initial data frame consisted of 1278 LIF spectra, with each spectrum containing between 3622 and 3653 variables. In principle, the band of the fluorescence signal is longer than the wavelength of the laser (405 nm in this case). Therefore, during the preprocessing of the fluorescence spectral data, only the 420–800 nm band of each fluorescence spectrum was retained, and the wavelengths that did not provide relevant information were discarded, resulting in 1871 variables per spectrum. This step was applied uniformly across all analyses. For Difference Spectra Analysis (DSA) and One-way Analysis of Variance (ANOVA), no further preprocessing was required, and the spectral data within the 420–800 nm range were used directly. For the PCA, machine learning, and deep learning models, additional preprocessing steps were implemented. The data were first restructured by pivoting to create a suitable format for analysis. A Standard Scaler was then applied to standardize the feature set, ensuring that all features contributed equally to the analysis by transforming them to have a mean of zero and a standard deviation of one. These preprocessing steps were consistently applied to all models to evaluate their performance on LIF spectral data with minimal manipulation.

2.4. Difference spectra analysis (DSA)

DSA is a spectral analysis method that offers detailed information about spectral changes, specifically to determine the boundaries of the main spectral components, as well as the direction of change of their susceptibility to external factors (Oparin, 2012).In this method, the difference spectra are derived by subtracting a reference spectrum from each spectrum within the group. The reference spectrum may vary, such as being the initial or final spectrum of the group, or the averaged spectrum, based on the intended information extraction. In this DSA implementation, for each angle, the mean spectrum of pure VOO detected at that angle serves as the reference. Difference spectra are then calculated by subtracting this reference from the mean spectra of VOO samples with different adulteration levels at the same angle.

To quantify overall sensitivity to adulteration at each angle, we sum the absolute values of these spectral differences across all adulteration levels. This sum represents the total spectral change due to adulteration, with higher values indicating greater sensitivity at that angle. The resulting sensitivity plot displays these summed differences for each detection angle, providing a visual representation of which angles are most sensitive to detecting adulteration in VOO samples. This approach allows for a direct comparison of how the spectral characteristics change with increasing adulteration at each specific angle, providing insights into the sensitivity of different detection angles to adulteration.

2.5. One-way analysis of variance (ANOVA) of LIF spectra

To assess the impact of detection angle on the LIF spectra across different adulteration percentages, we conducted a one-way ANOVA, a robust and widely used statistical method for group comparisons, for each wavelength and adulteration level. The spectral data were grouped by detection angle, and F-values were calculated to quantify the between-group variability relative to within-group variability (Freund and Wilson, 2003). This analysis was performed for each adulteration percentage (0 %, 20 %, 33.3 %, 42.8 %, 50 %, and 100 %) across all wavelengths (420–800 nm). The resulting F-values were plotted against wavelength for each adulteration percentage to visualize the spectral regions where angle-dependent variations were most pronounced. Additionally, we calculated the percentage of wavelengths showing statistically significant differences ($p < 0.05$) between angles for each adulteration level and overall. This approach allowed us to systematically evaluate how the influence of detection angle on spectral measurements varied with both wavelength and adulteration percentage.

2.6. Principal component analysis (PCA)

PCA, an unsupervised dimensionality reduction technique, was employed to visualize and analyze the complex spectral data obtained from the LIF measurements (Jolliffe, 2002). The intensity values across all wavelengths (420–800 nm) for each sample, adulteration percentage, and detection angle were used as input variables, allowing for visual assessment of clustering patterns and separability between different adulteration levels and detection angles. The PCA reduced the high-dimensional spectral data to three principal components (PCs). These PCs, representing the directions of maximum variance in the data, were used to create both 2D and 3D scatter plots. The first three PCs were chosen as they captured the most significant variations in the data, providing a comprehensive yet simplified representation of the spectral characteristics across different adulteration percentages and detection angles.

2.7. Machine learning classification algorithms

In this study, five machine learning techniques were selected for the classification of the detailed LIF spectral data obtained from eight different angles, aiming to develop robust models capable of multi-class discrimination among six adulteration levels. The machine learning models employed in this study include LDA, Partial Least Squares Discriminant Analysis (PLS-DA), SVM, Random Forest, and Extreme Gradient Boosting (XGBoost). All machine learning models were implemented using the Python library scikit-learn (Pedregosa et al., 2011).

2.7.1. Linear discriminant analysis (LDA)

LDA maximizes the ratio of between-class variance to within-class variance by fitting class-conditional densities to the data using Bayes' rule, resulting in a classifier with a linear decision boundary (Tharwat et al., 2017). The LDA model assumes that all classes share the same variance-covariance matrix and fits a Gaussian probability density to each class. In this study, the LDA model was implemented with optimized parameters to enhance its performance on the high-dimensional spectral data. The best-performing configuration utilized a shrinkage parameter of 0.5 and the LSQR solver. Shrinkage helps to improve the stability of the covariance matrix estimation, particularly useful in high-dimensional settings, while the LSQR solver is efficient for large

datasets and can handle potential multicollinearity in the features.

2.7.2. Partial least squares discriminant analysis (PLS-DA)

PLS-DA is a linear multivariate classification technique derived from Partial Least Squares regression (Ballabio and Consonni, 2013). PLS-DA performs dimension reduction on the predictor variables by extracting components designed to exhibit maximal correlation with the class membership (Van Ruth et al., 2010). A cross-validation was performed across a range of components (from 1 to 20) to determine the optimal number of components for the PLS-DA model. The number of components yielding the highest cross-validation score is selected as the optimal configuration, which in this case resulted in eight components. The PLS-DA model is then trained on the training set using the optimal number of components. Predictions are made on the test set, and these predictions are rounded and clipped to ensure they fall within the valid range of class labels.

2.7.3. Support vector machines (SVM)

Support Vector Machines are powerful supervised learning models used for classification and regression tasks (Cortes and Vapnik, 1995). In this study, a multi-class SVM classifier was implemented using scikit-learn's Support Vector Classification (SVC) class, which employs a One-vs-One approach for multi-class classification (Debnath et al., 2004). For the model's structure, a linear kernel was used, which is effective for high-dimensional data such as the spectral dataset in this study. The linear kernel was chosen for its efficacy with high-dimensional data, allowing for efficient classification while maintaining the flexibility of the standard SVM formulation.

2.7.4. Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees and combines their outputs for classification or regression. The model builds each tree using bootstrap sampling and random feature selection. For each tree, a subset of the training data is randomly sampled, and at each node, a random subset of features is considered for splitting. This process creates a diverse 'forest' of trees. For predictions, the model aggregates outputs from all trees, using majority voting for classification or averaging for regression. This ensemble approach, combining bagging (bootstrap aggregating) and random feature selection, enhances the model's generalization capability and reduces overfitting (Mansour and Schain, 2001). In this study, we optimized the Random Forest model by tuning several hyper parameters. The optimal configuration was found to use 200 trees (n_estimators), with no maximum depth limit (max_depth: None), a minimum of two samples required to split an internal node (min_samples_split), and a minimum of one sample required to be at a leaf node (min_samples_leaf).

2.7.5. Extreme gradient boosting (XGBoost)

XGBoost is an advanced implementation of gradient boosting for supervised learning tasks (Chen and Guestrin, 2016). It builds an ensemble of decision trees in a sequential manner, where each tree attempts to correct the errors of the previous ones (Fan et al., 2018). In this work, the following XGBoost parameters were set: n_estimators= 100 (number of boosting rounds), learning_rate= 0.1 (step size shrinkage used to prevent overfitting), min_child_weight= 5 (minimum sum of instance weight (hessian) needed in a child), gamma= 0 (minimum loss reduction required to make a further partition on a leaf node), subsample= 0.8 (subsample ratio of the training instances), and alpha= 0 (L1 regularization term on weights). These parameters were chosen to balance the model's complexity and performance.

2.8. Deep learning classification algorithms

In addition to machine learning methods, two deep learning models were implemented for the classification task: a Multilayer Perceptron (MLP) and a one-dimensional Convolutional Neural Network (1D-CNN). Both deep learning models were implemented using TensorFlow (Abadi et al., 2016) and Keras libraries, and were compiled with the Adam optimizer (Kingma and Ba, 2014).

2.8.1. Multilayer perceptron (MLP)

An MLP neural network consists of three main components: an input layer, one or more hidden layers, and an output layer. The input layer receives data, hidden layers process it, and the output layer produces the final result. Neurons in each layer are typically fully connected to those in adjacent layers, allowing the network to learn complex patterns and relationships in the data (Taud and Mas, 2018). The MLP model implemented in this study consisted of an input layer, two hidden layers with 64 and 32 neurons respectively, each followed by a dropout layer with a rate of 0.5 to prevent overfitting, and an output layer with six neurons corresponding to the six adulteration levels. The MLP model used sparse categorical cross-entropy as the loss function.

2.8.2. One-dimensional convolutional neural network (1D CNN)

The 1D-CNN model comprised two convolutional layers with 32 and 64 filters respectively, each followed by max pooling layers, a flatten layer, a dense layer with 128 neurons, a dropout layer with a rate of 0.5, and an output layer with six neurons. The 1D-CNN structure employs multiple convolution kernels to scan the input vector, forming feature maps in the convolutional layers, followed by pooling to compress the data, and finally a fully connected layer to output classification results (Chen et al., 2022). For the 1D-CNN, input data was reshaped to include a channel dimension, transforming it into a 3D array suitable for 1D convolution operations. Additionally, the 1D-CNN incorporated a learning rate scheduler that reduced the learning rate from an initial value of 0.001 as training progressed, to fine-tune the model's performance. The 1D-CNN used categorical cross-entropy as the loss function. Both MLP and 1D-CNN models used Rectified Linear Unit (ReLU) activation for hidden layers and softmax (normalized exponential function) for the output layer.

2.9. Model evaluation

To ensure robust evaluation of the classification models, and prevent overfitting, we employed k-fold cross-validation with k = 10 (Berrar, 2019). The dataset was randomly split into ten equal-sized subsamples. For each fold, one subsample was retained as the validation set, while the remaining nine were used for training. This process was repeated 10 times, with each subsample used exactly once as the validation set. Performance metrics including accuracy, precision, recall, and F1-score were calculated for each fold and then averaged to obtain the overall performance of each model. Additionally, predictions from each fold were aggregated to create a final confusion matrix, providing a comprehensive view of the model's classification performance across all folds.

2.10. Cross-validation across different angles

The study implemented novel cross-validation strategies, such as Single-Angle Cross-Validation (SACV) and Leave-One-Angle-Out Cross-Validation (LOAOCV), to explore the generalizability of the models to effects of the sIFE on LIF spectra across various detection angles. These methods were specifically developed to investigate the impact of sIFE on classification accuracy and to evaluate the ability of models trained on data from single specific or multiple detection angles to classify samples measured at unseen detection angles. This involved systematically training and testing the models with different combinations of data to simulate real-world variability and assess robustness.

The experimental cross-validations were conducted using three of the best-performing models from the initial analysis: SVM, Random Forest, and 1D-CNN. These models were selected based on their superior performance in the initial classification tasks and their diverse

approaches to machine learning. The structures of these models were implemented consistently with those described in Section 2.7.3 (SVM), 2.7.4 (Random Forest), and 2.8.2 (1D-CNN).

2.10.1. Single-angle cross-validation (SACV)

The SACV method was implemented to evaluate the performance of models trained on LIF spectral data collected from one specific detection angle and tested on data from the same angle, as well as their ability to generalize to data from different, unseen detection angles. The dataset was first split by the detection angles, creating separate subsets for each angle. To ensure a fair comparison, the sample sizes were standardized across all angles to the minimum number of samples available for any single angle. The data from each angle was then split into 80 % for training and 20 % for evaluation. For each iteration, the training data from one specific angle was selected as the training set. The model was then trained exclusively on this training set from the selected angle. Once trained, the model was tested on the test data from all other detection angles, as well as the held-out validation set from the same detection angle as the training set. This process was repeated for all eight detection angles (20°, 30°, 40°, 50°, 60°, 70°, 80°, and 90°). This specialized cross-validation provides insights into the impact of sIFE on the conventional approach, where models are typically trained and tested using LIF data obtained from a single, specific angle, by revealing how accuracy changes when training and testing angles differ.

2.10.2. Leave-one-angle-out cross-validation (LOAOCV)

The LOAOCV method was employed to evaluate the performance of models trained on LIF data collected from multiple detection angles. This method provides assessment of the generalization capabilities of these models by ensuring that the training set includes data from various angles, while the test set consists of data from a completely unseen angle. This approach is specifically adapted to account for different detection angles in our dataset. For each iteration, data from one specific angle was left out as the test set, while data from all other angles was used for training. The model was trained on this multi-angle training set and then tested on the left out angle. This process was repeated for all eight detection angles, ensuring that each angle was used as the test set once. The LOAOCV method allows for an evaluation of how well the models trained on data from multiple angles can generalize to data from new, unseen angles.

2.11. Overview of the analysis framework

The secondary inner filter effect, resulting from self-absorption of the fluorescence emission, leads to spatial heterogeneity in the fluorescence signals, causing the spectral data to vary with the angle of detection. The angular dependence of LIF spectral features, attributed to spatial anisotropy in the sIFE [29], indicates that inner filter effects vary with the detection angle and the composition of the SO adulterant in VOO. This study employed novel implementations of various analytical methods to investigate these angular variations and address the unique challenges posed by the angular dependence of LIF spectra in quantitative adulteration detection.

The methodological approach ranged from spectral analysis techniques to advanced machine learning and deep learning classification models. DSA was adopted to quantify spectral changes across different angles, while One-way ANOVA provided statistical insights into the angular variations. Principal Component Analysis (PCA) facilitated dimensionality reduction and visualization of spectral variations across angles and adulteration levels. The classification task utilized a variety of machine learning algorithms, including linear methods such as LDA and PLS-DA, non-linear approaches like SVM, and ensemble methods such as Random Forest and XGBoost. Deep learning models, including MLP and 1D-CNN, were also implemented to leverage their capacity for automatic feature extraction and complex pattern recognition using LIF spectral data. Additionally, novel experimental cross-validation strategies were used to assess the feasibility of a new classification approach with the goal of eliminating the detection setup limitations in the LIF setup.

While this study introduces innovative applications of these analytical methods in the context of LIF spectroscopy for VOO adulteration with SO, it's important to note that these techniques have already been widely exploited in similar areas of variation analysis and food adulteration classification using spectral data. Table 1 provides a summarized overview of the analytical methods and their specific applications in the present study. Importantly, by referencing examples of implementations of methods in previous research, the table highlights both the novelty of our approach and the established utility of these analytical techniques in related research areas, serving to contextualize our methodological approach within the broader scientific literature.

3. Results and discussion

3.1. Spectra analysis

Fig. 1 shows representative LIF spectra of Virgin olive oil (VOO), Sunflower oil (SO), and their various mixture ratios, recorded with an excitation wavelength of 405 nanometers (detected at angle θ=20° with respect to the laser beam). For VOO, a peak at the wavelength of 685 nanometers is observable due to the presence of chlorophyll. When SO is added, this peak decreases due to the reduction in the mass percentage of chlorophyll. However, at a wavelength of 520 nanometers, an increase in the peak is observed in this region due to the rise in carotenoids in the mixture of the two oils.

Crucial insights into the sIFE in OO adulteration detection have been provided by the investigation of the angular dependence of anisotropy in LIF spectral features, as reported by Bavali et al. (2025). It was demonstrated that LIF intensity decreased as the detection angle increased, revealing the spatial anisotropy caused by sIFE. The angular dependence of anisotropy in LIF spectral features was observed to vary with the volume ratio of SO in VOO and the specific detection angle with respect to the direction of the laser beam used to collect the LIF spectra. In the present study, the DSA was employed to investigate the sensitivity of different detection angles to VOO adulteration with SO utilizing the LIF spectra collected at various detection angles relative to the laser beam direction. Fig. 2.a illustrates the total spectral difference for each detection angle, representing the overall sensitivity to adulteration. The results reveal a clear trend in the sensitivity of different detection angles to VOO adulteration. The most striking observation is the significantly higher sensitivity at the 20° detection angle, with a total spectral difference of 1862,551.70. This value is more than twice that of the next most sensitive angle (30°, with 870,547.45) and approximately six times higher than the least sensitive angle (90°, with 306,223.51). The sensitivity to adulteration demonstrates a consistent decrease as the detection angle increases from 20° to 90°. This trend can be attributed to the sIFE, which becomes more pronounced at larger angles. At smaller angles, such as the 20° detection angle where the sensitivity is highest, the fluorescence encounters fewer non-excited molecules before reaching the detector. As the detection angle increases, the optical path traverses a larger volume of the sample, encountering more non-excited chlorophyll-a, pheophytin-a, and pheophytin-b molecules within the field of view of the fiber probe (Khayatian et al., 2023).

The F-values of the one-way ANOVA analysis, illustrated in Fig. 2.b, reveal significant insights into the angular dependence of LIF spectra across different adulteration levels. The ANOVA results further underscore the significant impact of detection angle on LIF spectra across different adulteration levels. For VOO (0 % adulteration), 81.83 % of the wavelengths showed significant differences between angles, indicating a strong angular dependence even in unadulterated samples. As the adulteration level increased, the percentage of wavelengths with significant angular differences also increased, reaching 100 % for adulteration levels of 20 % and above. Notably, SO (100 % adulteration)

Table 1
Overview of Analytical Methods Employed for Spectral Analysis, Multi-Class Classification, and Model Generalizability Assessment Using the 1278 LIF Spectra from Six Adulteration Levels at Eight Detection Angles (θ).

Analytical Methods	Spectral Analysis of Angular Variations	Classification of Multi-Angle LIF Spectra	Evaluation of Model Generalizability Across Detection Angles	Prior Implementations
DSA	√	×	×	-
ANOVA	√	×	×	(Zhang et al., 2021, Lu et al., 2018)
PCA	√	×	×	(Zhang et al., 2011, Firouz et al., 2022)
LDA	×	√	×	(Stefas et al., 2021)
PLS-DA	×	√	×	(Hirri et al., 2016, Vieira et al., 2021)
SVM	×	√	√	(Zheng et al., 2020)
Random Forest	×	√	√	(Tian et al., 2023)
XGBoost	×	√	×	(Gomes et al., 2022)
MLP	×	√	×	(Torreblanca-Zanca et al., 2019)
1D-CNN	×	√	√	(Zhao et al., 2022)

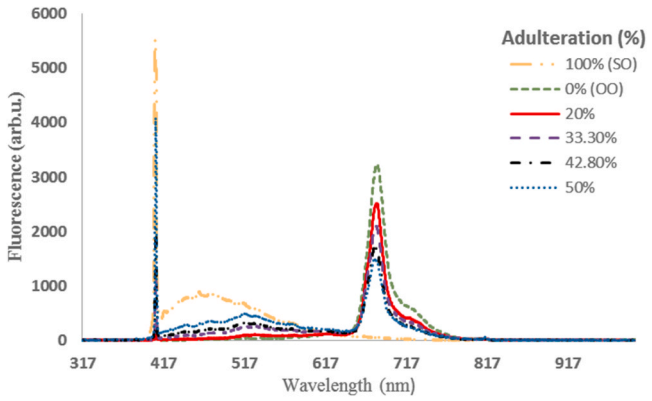


Fig. 1. Typical LIF spectrum of Oil Samples Containing Different volume ratios of VOO and Sunflower oil (SO) excited by laser at 405 nm and recorded at angle θ= 20°.

showed significant angular differences in 96.69 % of wavelengths, slightly lower than the fully adulterated VOO samples but still substantially higher than VOO. This trend suggests that the introduction of SO into VOO enhances the angular dependence of the LIF spectra across a broader range of wavelengths. The overall statistics, showing that 89.90 % of wavelengths exhibit significant differences between angles across all samples, emphasize the pervasive influence of detection angle on LIF spectral characteristics.

The F-values, plotted against wavelength for each adulteration percentage, demonstrate distinct patterns and variations in the spectral regions where angle-dependent differences are most pronounced. For VOO, the F-value plot shows a unique pattern compared to adulterated samples. There is a notable peak in the F-values around 650–700 nm, reaching values above 10^4. This suggests that the detection angle has a particularly strong influence on the LIF spectra of VOO in this wavelength range, which corresponds to the fluorescence emission of chlorophyll and its derivatives. The high F-values indicate significant variations in spectral intensity across different detection angles, emphasizing the importance of angular considerations in VOO analysis. In contrast, the SO sample exhibits consistently high F-values (around 10^3) across a broad range of wavelengths (450–650 nm). This indicates that the angular dependence of LIF spectra is strong and relatively uniform across this spectral region for SO. The absence of a pronounced peak suggests that SO's spectral properties lack significant overlap between the absorption spectra of SO and its fluorescence spectra, which results in the absence of sIFE. The adulterated VOO samples show intermediate patterns between VOO and SO. As the adulteration percentage increases, there is a general trend of increasing F-values, particularly in the 450–650 nm range. This suggests that the angular dependence of the LIF spectra in this region becomes more pronounced as adulteration percentage increases. The 650–700 nm region shows high angular dependence for VOO, as evidenced by the prominent peak in the F-value for the VOO sample. This peak becomes less pronounced in adulterated samples as the adulteration percentage increases, potentially offering a specific marker for VOO purity. Conversely, the 450–650 nm region shows a gradual increase in F-values with increasing adulteration percentage. The contrasting behavior in these two spectral regions provides a potential means for quantifying the degree of adulteration, as the angular dependence of the LIF spectra shifts from the characteristics of VOO to those of SO. It is noteworthy that the regions below 450 nm and above 750 nm generally show lower F-values, indicating less angular dependence in these spectral ranges. This suggests that these wavelengths might be less informative for this adulteration detection when considering angular variations. These findings emphasize the need for careful consideration of detection angles and wavelength ranges in developing robust LIF spectroscopy methods for quantitative analysis of oil adulteration. The complex interplay between adulteration level, wavelength, and angular dependence underscores the importance of a multi-faceted approach in VOO authentication.

3.2. PCA results

PCA was chosen as an initial analysis method due to its ability to reduce the dimensionality of complex spectral data while preserving the most important variations, allowing for effective visualization and preliminary assessment of the dataset's structure. PCA was performed on the LIF spectral data to visualize and analyze the complex relationships between oil compositions, detection angles, and spectral characteristics. The PCA revealed that the first three principal components (PCs) account for a cumulative 96.18 % of the total variance in the dataset. Specifically, PC1 explains 62.23 % of the variance, PC2 accounts for 30.71 %, and PC3 contributes an additional 3.23 %. This high cumulative explained variance indicates that these three PCs capture the majority of the relevant spectral information.

Fig. 3.a, showing the 2D PCA plot which represents 92.94 % of the total variance, reveals distinct clustering patterns for different oil compositions. VOO samples (green markers) and SO samples (yellow markers) form well-separated clusters, indicating clear spectral differences between these two oils. The adulterated samples (20 %, 33.3 %, 42.8 %, and 50 %) show a gradual transition between these two extremes, with increasing adulteration levels generally moving from left to right along PC1. This suggests that PC1, which accounts for the largest portion of variance, primarily captures the spectral changes associated with increasing SO content. The plot demonstrates that samples with different adulteration levels are largely separable. However, some overlap is observed between adjacent adulteration levels, particularly in

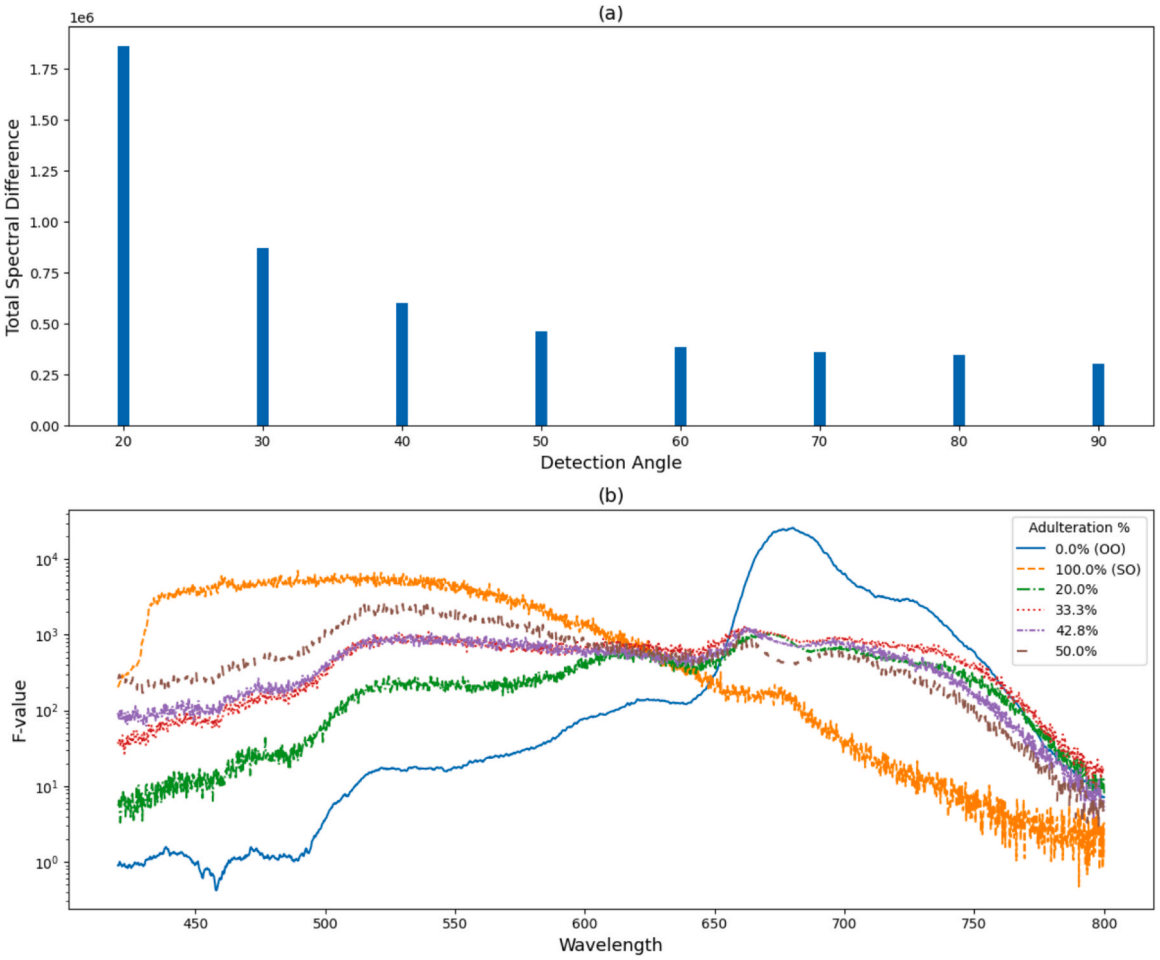


Fig. 2. (a) Overall Sensitivity to Adulteration for Each Detection Angle Based on DSA. (b) One-Way ANOVA F-values Across Wavelengths for Different Levels of VOO Adulteration with Sunflower Oil (SO).

the mid-range (33.3–50 %), indicating that fine discrimination between these levels may be challenging.

The 3D PCA plot shown in Fig. 3.b includes PC3, providing a representation that accounts for 96.18 % of the total variance. This additional dimension enhances the visualization of cluster searations and the influence of detection angles. The 3D plot confirms the general trends observed in the 2D plot while offering a more comprehensive view of the data structure. The influence of detection angle on the PCA results is evident from the spread of data points within each oil and adulteration level. For each oil composition, data points form elongated clusters, with different markers representing different detection angles. This spread indicates that the detection angle contributes significantly to the spectral variations captured by the PCs. The influence of detection angle is more pronounced in the 3D plot, with data points for each adulteration level forming more complex, three-dimensional distributions. The PCA results demonstrate the potential of LIF spectroscopy combined with multivariate analysis for detecting and quantifying VOO adulteration with SO. Overlaps between some adulteration levels and the significant influence of detection angle highlight the need for sophisticated classification algorithms. This unsupervised analysis sets the stage for further exploration using machine learning and deep learning techniques, which may exploit the subtle spectral differences and angular dependencies to achieve more accurate adulteration quantification.

3.3. Classification analysis

The performance of various machine learning and deep learning models in classifying different oil compositions was evaluated using multiple metrics, as summarized in Table 2. Among the traditional machine learning approaches, SVM and Random Forest demonstrated exceptional performance. The SVM model achieved perfect scores across all metrics (accuracy, precision, recall, and F1-score of 1.0), indicating its robust capability in distinguishing between different adulteration levels. The Random Forest model closely followed with an average accuracy of 99.61 %, showcasing the effectiveness of ensemble methods in handling the complex spectral data. XGBoost also performed admirably, with an average accuracy of 99.14 %, further reinforcing the strength of tree-based ensemble methods for this classification task.

In contrast, LDA and PLS-DA showed comparatively lower performance, with accuracies of 89.67 % and 81.30 %, respectively. This suggests that the relationship between spectral features and adulteration levels may be too complex for these linear methods to fully capture. The superior performance of non-linear models like SVM and Random Forest indicates that the classification task involves intricate, non-linear relationships in the spectral data, which these models are better equipped to handle. In the realm of deep learning, both the MLP and the 1D-CNN exhibited impressive performance. The MLP achieved an impressive average accuracy of 98.59 %, while the 1D-CNN matched the SVM's perfect performance across all metrics. The superior performance of the 1D-CNN underscores the effectiveness of convolutional architectures in processing spectral data, likely due to their ability to capture local patterns and spatial relationships within the spectra. This suggests that the 1D-CNN is particularly well-suited for analyzing the complex, multi-angle LIF spectral data used in this study. Overall, these results

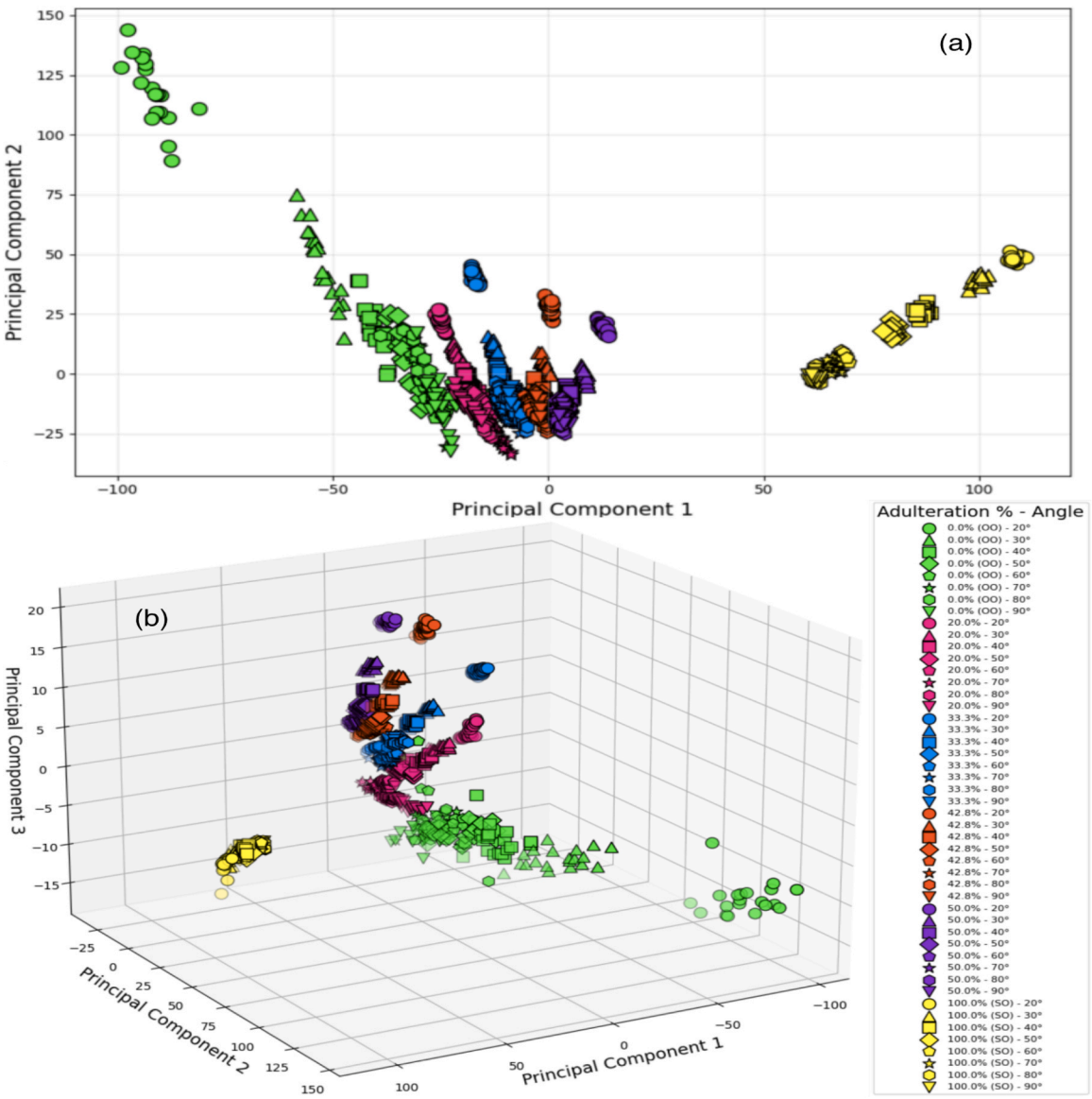


Fig. 3. 2D (**a**) and 3D (**b**) PCA Plots of LIF Spectral Data Showing Clustering Patterns of Oil Compositions recorded at various detection angles with respect to the laser beam direction.

Table 2
Performance Metrics of Machine Learning and Deep Learning Models in Multi-Class Classification of Olive Oil Adulteration Using the 1278 LIF Spectral Data obtained from six adulteration levels and eight detection angles. Accuracy, precision, recall, and F1-scores are calculated for each iteration of the k-fold cross-validation evaluation and averaged.

Classification Model	Accuracy	Precision	Recall	F1-Score
LDA	0.897	0.909	0.911	0.907
PLS-DA	0.813	0.837	0.822	0.817
SVM	1.000	1.000	1.000	1.000
Random Forest	0.996	0.997	0.996	0.996
XGBoost	0.991	0.993	0.990	0.991
MLP	0.986	0.989	0.987	0.987
1D-CNN	1.000	1.000	1.000	1.000

demonstrate the high potential of advanced machine learning and deep learning techniques in accurately classifying different oils and adulteration levels using multi-angle LIF spectral data. The near-perfect performance of models like SVM, Random Forest, and 1D-CNN suggests that these approaches can effectively capture the complex relationships between spectral features, detection angles, and adulteration levels. The significant performance gap between linear and non-linear models highlights the complexity of the classification task and the importance of choosing appropriate modeling techniques for spectroscopic data analysis in food authentication applications.

3.4. *Single-angle cross-validation (SACV)*

While Section 3.3 evaluated model performance using standard k-fold cross-validation on the full dataset (1278 spectra across all angles and adulteration levels), Sections 3.4 and 3.5 focus on specialized validation strategies to assess angular generalizability. The same dataset is utilized across all sections, but Sections 3.4 and 3.5 implement novel cross-validation protocols (SACV and LOAOCV) that systematically partition the data by detection angle rather than random sampling. This angular partitioning isolates the impact of sIFE on model performance, enabling direct comparison of single-angle versus multi-angle training paradigms. Unlike Section 3.3, which evaluates classification accuracy across all angles simultaneously, Sections 3.4–3.5 quantify how models generalize to unseen detection geometries, a critical distinction for real-

world applications.

The SACV analysis was conducted on three of the best-performing models: SVM, Random Forest, and 1D-CNN. This analysis aimed to evaluate how the sIFE impacts the accuracy of the conventional method of training models on data from a single detection angle. The SACV method was employed to train each model on data from one specific angle and test it on data from both the same angle and different, unseen detection angles. This approach helps assess the impact of sIFE on model accuracy and identify detection angles that offer the highest generalizability. The results of this analysis are presented in Fig. 4, which illustrates the performance of each model across different training and testing detection angle combinations.

Fig. 4.a displays the performance of the SVM model. When trained and tested on data of the same angle, the SVM consistently achieves perfect accuracy (1.00) across all angles. However, a significant drop in accuracy was noted when the model was tested on data from different detection angles, achieving an off-diagonal average accuracy of 0.722. The Random Forest model in Fig. 4.b, exhibited a similar trend, with high accuracy when trained and tested on the same angle, achieving a diagonal average accuracy of 0.996. However, considerable drops in accuracy were observed when tested on data from other detection angles, with an off-diagonal average accuracy of 0.610. Notably, the Random Forest model trained on data from the 20° angle showed significantly lower accuracy when tested on data from other angles. The 1D-CNN model, illustrated in Fig. 4.c, shows a pattern similar to the SVM and Random Forest models, achieving perfect accuracy when trained and tested on the same angle, with a diagonal average accuracy of 1.000. However, its generalization performance varies considerably across different training and testing detection angle combinations. Specifically, the model trained on the 20° angle performed poorly on other angles, whereas models trained on higher angles, such as 60° and 90°, exhibited better generalizability. The off-diagonal average accuracy for the 1D-CNN model was 0.619, indicating that while it performs well on the same angle, its generalization to different detection angles is limited. In summary, the SACV analysis revealed that while conventional single-angle training methods can achieve high accuracy with the models, the classification performance is significantly impacted by sIFE when tested on data from different detection angles.

3.5. *Leave-one-angle-out cross-validation (LOAOCV)*

The LOAOCV analysis was conducted to evaluate the generalization capabilities of models trained on LIF data collected from multiple detection angles. This method ensures that the training set includes data from various angles, while the test set consists of data from a completely unseen angle.

The results of the LOAOCV analysis are presented in Fig. 5, which illustrates the accuracy of each model when tested on different held-out angles. The SVM model (Fig. 5.a) demonstrated remarkable consistency, achieving perfect accuracy (1.00) for all angles except 80°, where it still maintained a high accuracy of 0.956. This result underscores the SVM model's robustness in generalizing across different detection angles. The Random Forest model (Fig. 5.b) also showed strong performance but with more variability compared to the SVM. The Random Forest model achieved perfect accuracy (1.00) when tested on the 40°, 50°, and 60° angles, and near-perfect accuracy on the 70° and 80° angles, with 0.981 and 0.925, respectively. However, its performance dropped significantly when tested on the 20° angle, with an accuracy of 0.667, and to a lesser extent on the 90° angle, with an accuracy of 0.894. These results suggest that while the Random Forest model is generally effective, its ability to generalize to certain detection angles, particularly smaller ones like 20°, is less dependable. The 1D-CNN model (Fig. 5.c) exhibited high performance across most angles but showed notable variability. The model achieved perfect accuracy (1.00) for several angles, including 30°, 40°, and 60°, and near-perfect accuracy for angles like 50°, 70°, and 90°, with accuracies of 0.981, 0.981, and 0.994, respectively. However, the model's accuracy dropped substantially when tested on the 20° angle, achieving only 0.585. This indicates that while the 1D-CNN model is highly effective at most angles, its performance is significantly weakened when testing on the 20° detection angle. Overall, the LOAOCV analysis highlights the strengths and limitations of each model in handling data from multiple detection angles. The SVM model demonstrated the highest consistency and robustness, making it a strong candidate for applications requiring reliable generalization across various angles. The Random Forest and 1D-CNN models also performed well but showed more variability, particularly at smaller angles like 20°. These findings emphasize the importance of training models on data from multiple detection angles to enhance their generalization capabilities and mitigate the effects of sIFE.

The concept of the LOD interval establishes a dynamic detection range between the minimum LOD, which represents the smallest concentration distinguishable from zero, and the maximum LOD, the highest concentration detectable within the interval's limits. Further details on the calculation of these parameters are available in (Allegrini and Olivieri, 2016). Based on the data recorded at a 20-degree angle, the calculated minimum and maximum LOD values were 0.046249 % and 0.334172 %, respectively.

To further validate and extend the practical applicability of the

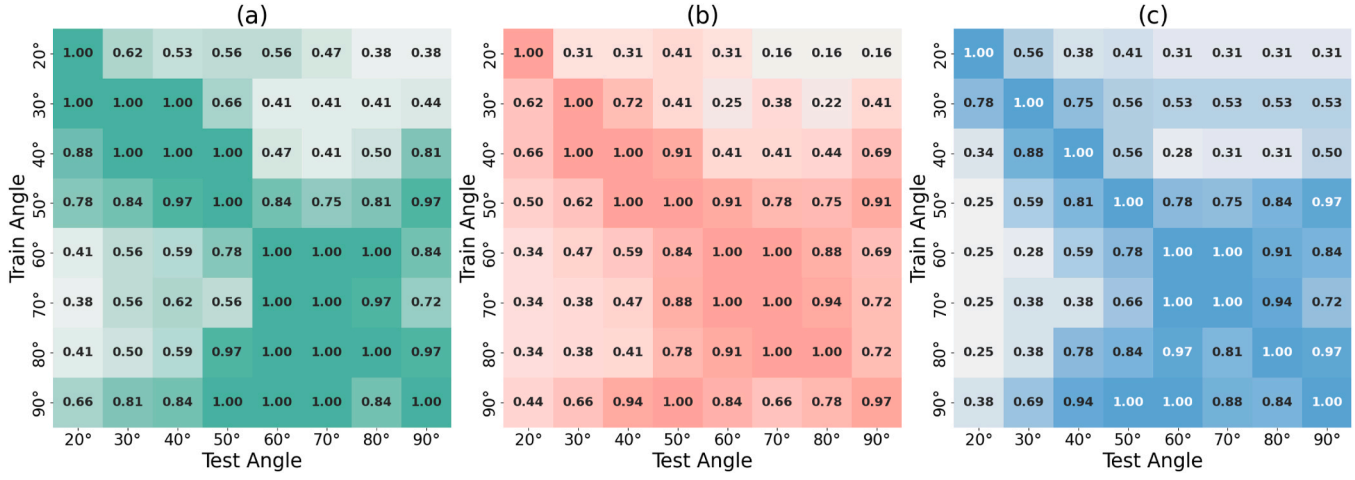


Fig. 4. Heatmaps Showing Classification Accuracies of (a) SVM, (b) Random Forest, and (c) 1D-CNN Models in Single-Angle Cross-Validation (SACV). The models were trained using LIF spectral data collected at a specific detection angle (Train Angle) and evaluated using other data recorded at various detection angles (Test Angle). Darker shades in each heatmap indicate higher accuracy values, approaching one.

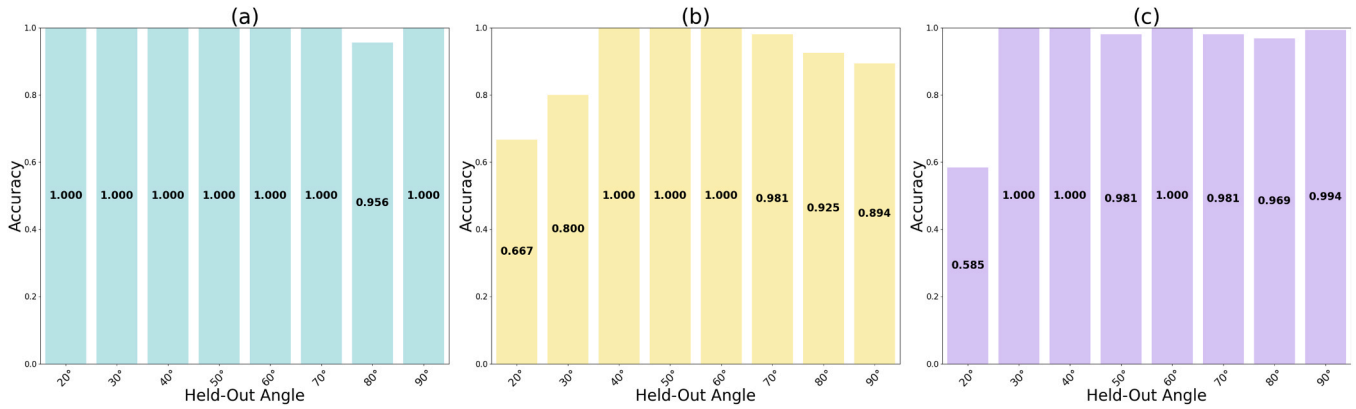


Fig. 5. Leave-One-Angle-Out Cross-Validation (LOAOCV) for (a) SVM, (b) Random Forest, and (c) 1D-CNN Models. Bars represent accuracy when LIF spectral data from each detection angle is held out as the test set, with the training set comprising data from all other angles.

proposed methodology, future studies could explore its implementation across larger-scale datasets encompassing diverse adulterants, oil varieties, and setup geometries. Expanding the methodology to encompass a wider array of edible oils, adulteration scenarios, and spectral acquisition setups would provide valuable insights into its robustness and generalizability. Such efforts would not only reinforce the reliability of the approach but also facilitate its adaptation to broader food authenticity challenges, where variations in composition, fluorescence properties, and measurement configurations are inherent.

4. Conclusion

This study demonstrates the effectiveness of combining LIF spectroscopy with machine learning and deep learning techniques to detect and quantify the adulteration of VOO with SO. By addressing sIFE and utilizing spectral data from multiple detection angles, we developed robust models for accurate classification. Difference Spectra Analysis (DSA) revealed that smaller detection angles, especially at 20°, were most sensitive to fluorescence changes, highlighting the significant impact of sIFE on LIF spectra. One-way ANOVA validated strong angle-dependent variations in the spectral data, identifying key regions affected by detection angle changes. Principal Component Analysis (PCA) effectively visualized the complex spectral data, enhancing our understanding of the relationships among oil compositions and detection angles. Our machine learning and deep learning models, including SVM, Random Forest, and 1D-CNN, achieved high classification accuracies, showcasing their ability to exploit subtle spectral variations. Cross-validation strategies, such as Single-Angle Cross-Validation (SACV) and Leave-One-Angle-Out Cross-Validation (LOAOCV), provided insights into model generalizability, revealing that models trained on multiple angles performed better on unseen data compared to those trained on a single angle. The findings of this research have significant implications for food quality authenticity analysis, offering a methodology that enhances the accuracy and generalizability of LIF spectroscopy. By mitigating the limitations imposed by sIFE and enabling the use of spectral data from various angles, our approach simplifies the implementation of LIF spectroscopy, making it a practical and versatile tool for ensuring the authenticity and quality of olive oil and other edible oils.

CRediT authorship contribution statement

Chegini Reza: Writing – original draft, Visualization, Software, Methodology. **Bavali Ali:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Data curation, Conceptualization. **Rahmatpanahi Ali:** Writing – original draft, Visualization, Resources, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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